

Daily Problem: 13–Feb–2026

Problem Statement

Let Σ be a finite alphabet and let r^c denote the *complement* of a regular expression r , i.e.

$$L(r^c) = \Sigma^* \setminus L(r).$$

Regular expressions built using only concatenation, union, and complement (no Kleene star) define the *star-free* languages.

Give star-free regular expressions for the following languages (assume $\Sigma = \{a, b\}$ unless stated otherwise):

- (i) $\{w \in \Sigma^* : w \text{ contains at least one } a\}$.
- (ii) $(ab)^* = \{(ab)^n : n \geq 0\}$.
- (iii) $\{w \in \Sigma^* : \text{between the first and last } a \text{ (if they exist) there is no } b\}$.

Solution

Throughout, we use only union ($\cup$), concatenation, and complement $(\cdot)^c$. We also freely use parentheses.

(i) Words containing at least one a

A word contains no a iff it is made only of b 's, i.e. it lies in b^* . But b^* itself uses star, so we express b^* as the complement of “contains at least one a ” instead.

The clean star-free expression for “contains at least one a ” is:

$$\boxed{((b^c))^c}$$

where b^c means $\Sigma^* \setminus \{b\}$. This is not yet what we want since it hides star.

A standard star-free way is to use the fact that

$$\text{“no } a\text{”} = \{b\}^* = (\Sigma^* a \Sigma^*)^c$$

and hence

$$\text{“at least one } a\text{”} = \Sigma^* a \Sigma^*.$$

So we want a star-free expression for Σ^* . In star-free syntax, we treat Σ^* as the universe language, which is allowed as a constant (equivalently $\emptyset^c$).

Thus:

$$\Sigma^* = \emptyset^c.$$

Therefore a star-free expression for “contains at least one a ” is

$$\boxed{(\emptyset^c) a (\emptyset^c)}.$$

(ii) $(ab)^*$

This is the language of all words that are *exactly* repetitions of the block ab , including ε .

A word is *not* in $(ab)^*$ if and only if at least one of the following holds:

- it starts with b ,
- it ends with a ,
- it contains aa as a substring,
- it contains bb as a substring.

Indeed, any repetition of ab alternates $a, b, a, b, \dots$ and must start with a (unless it is empty) and end with b (unless it is empty).

Let

$$U = \emptyset^c = \Sigma^*.$$

Then the four “bad” sets can be written star-free as:

$$B_1 = bU, \quad B_2 = Ua, \quad B_3 = UaaU, \quad B_4 = UbbU.$$

So the set of all bad words is $B_1 \cup B_2 \cup B_3 \cup B_4$.

If we take the complement, we get all alternating words that do not start with b and do not end with a , which includes ε and exactly $(ab)^*$.

Hence a star-free expression for $(ab)^*$ is:

$$\boxed{((bU) \cup (Ua) \cup (UaaU) \cup (UbbU))^c} \quad \text{where } U = \emptyset^c.$$

(iii) No b between first and last a

Interpretation: if the word has fewer than two a 's, the condition holds vacuously. If it has at least two a 's, then in the factorization

$$w = x a y a z,$$

where the first displayed a is the *first* a in w and the second displayed a is the *last* a in w , we must have that y contains no b , i.e. $y \in a^*$. Over $\{a, b\}$, that means y is a string of only a 's.

Equivalently, the language consists of all words that *do not* have a b occurring *between* two a 's. That forbidden pattern is exactly the substring $a(\text{something})b(\text{something})a$. In regex form, the bad set is:

$$\Sigma^* a \Sigma^* b \Sigma^* a \Sigma^*.$$

So the desired language is its complement:

$$\boxed{(U a U b U a U)^c} \quad \text{where } U = \emptyset^c.$$

Final Answers

Let $U = \emptyset^c = \Sigma^*$. Star-free regular expressions:

- (i) $\boxed{U a U}$.
- (ii) $\boxed{((bU) \cup (Ua) \cup (UaaU) \cup (UbbU))^c}$.
- (iii) $\boxed{(U a U b U a U)^c}$.